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Temporal variation in concentration of SOx and NOx in ambient air of Ranchi city

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Abstract : Atmospheric concentration of sulfur oxides and nitrogen oxides increases with growing population requirements. The stationary and mobile sources cause emission of SOx and NOx. A temporal variation is seen in the concentration of SOx and NOx. During the winter months the concentration is higher as compared to summer months.

Keywords: NOx, SOx, air pollution.

INTRODUCTION

Air pollution is a complex issue. It originates by sources including vehicular exhaust, industrial emissions, emissions from fossil fuels, construction activities to domestic activities. It has a deleterious effect on plants and animals, including human beings. In India air pollution is mainly caused from three sources: vehicles, industries, and domestic sources. At different spatial- temporal scales, worldwide systematic monitoring and analysis of the causes and effects of air pollution are being carried out. Different countries have defined standards for each criteria pollutant and have strict mechanisms for control of air pollution because clean air is a basic requirement for human health and well being.

Air pollution causes pernicious effects on human health, especially in areas with high population density. Forecasting air quality is one of the most important topics of research today for urban air pollution studies and specifically for prediction of pollution episodes.

The rapid industrialization leading to urbanization, unplanned and excessive exploitation of natural resources have been causing pollution problems in cities and towns of developing countries. Man made and natural sources of emissions have created toxic substances, which pollute the air. The problem of air pollution is affecting the urban environment of entire world.

Among the criteria pollutants, the effect of SOx and NOx has been studied widely. SOx and NOx are toxic pollutants. These are responsible for killing the aquatic life, damaging buildings and monuments, corroding copper and lead pipings, reducing soil fertility and causing toxic metals to leach into underground driving water.

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MATERIAL AND METHODS

2.1. DESCRIPTION OF STUDY AREA

Ranchi city, the capital of Jharkhand State lies at 23°22'N latitude 85°20'E longitude. Its municipal area is 175.12 square kilometers, and its average elevation is 651 m above sea level. It is located in the southern part of the Chotanagpur plateau, which is the eastern section of the Deccan plateau. Ranchi is known as the "City of Waterfalls" because of its numerous waterfalls, the most well known of which are 'Dassam', 'Hundru', 'Jonha', 'Hirni' Falls and 'Panchghagh' Falls. The topography of Ranchi is undulating. The hilly terrain is moderately covered with tropical forests. The climate of this city at one time was very salubrious and the city was accorded a 'hill station' status during the period of British rule. Since 1960s, rapid population growth and industrialization have caused a marked change in its weather patterns and an increase

in average temperature. This has resulted in gradual loss of its eligibility for "hill station" status. The city witnessed a sudden surge in population after the declaration of this part of the erstwhile Bihar state as a new state of Jharkhand in 2000. Owing to the rising employment opportunities and opening of numerous regional and state level offices, banks, and FMCG companies, the city witnessed a rapid influx of employment seeking migrants. As per latest 2011 India census, Ranchi city has a population of 1,073,440, making it the 37th largest urban city in India and third largest city in Jharkhand after Jamshedpur and Dhanbad. Ranchi is a city of industrious and enterprising people, drawn from all corners of Jharkhand and neighbouring states. There are several industries located in and around Ranchi city due to the precious mineral gift endowed to this part of the country by the nature. This city is well connected with other parts of the country. The National Highway 23, 33, and 75 passes through Ranchi city. Roads connecting major cities are providing movement for passenger and freight traffic. A new ring road is under construction around the city to reduce the growing traffic overload on the city roads. Bulk of the public transport in the city constitute of auto rickshaw and cycle rickshaw

2.2. SAMPLING SITES

For the present study three typical areas representing commercial zone, residential zone, and industrial zone were chosen for sampling of ambient air quality. Due to unplanned growth of the city it was very difficult to clearly demarcate any part of the city as exclusively commercial, residential, or industrial area. However, the three under mentioned parts of the city was selected for the present study.

Kantatoli Chowk: It is one of the busiest traffic intersections of the City. The National Highway 33 and the National Highway 23 intersects at this chowk and converges with the arterial roads of the city. Since no bypass is available, hundreds of heavy vehicles carrying goods also cross this intersection daily.

Ashok Nagar: This place is a developed residential part of the city. It is a posh area. Some banks and general stores are situated here. The development of this residential area has been done in a planned manner.

Tupudana: This area is located in fringe of the city. The industrial estate at Tupudana has several medium and small scale industries.

2.3. AMBIENT AIR QUALITY DATA

Ambient air quality was monitored on monthly basis between December 2010 and November 2011 at the above three sampling sites for criteria air pollutants SOx and NOx. The respirable dust sampler fitted with gaseous sampling attachment (APM 411) was used for monitoring of NOx and SOx values. The sampling for NOx and SOx was done using absorbing reagent. The samples of SOx and NOx were collected in glass impingers using potassium tetrachloromercurate(TCM) and absorbing solution of sodium hydroxide and sodium arsenite respectively. Oxides of sulfur were determined by Modified West & Gaeke method while oxides of nitrogen by modified Jacobs & Hochheiser Method.

RESULTS AND DISCUSSION

Nitrogen oxides are the major air pollutants in city atmospheres. The minimum value in Kantatoli was recorded in July (22.85µg/m^3) while the maximum in December (64.8µg/m^3). In Ashoknagar, the minimum concentration was seen in August (16.4 µg/m^3), while the maximum was recorded in February (37.16 µg/m^3). In Tupudana Industrial Area , the minimum values were obtained in February (29.32 µg/m^3) and the maximum in December (59.39 µg/m^3). Table 1 depicts the monthly variation in NOx values at three sampling sites. Increase in the concentration of nitrogen oxides in winter months has been reported by Hargreaves *et al* (2000), Reddy and Ruj (2003), and Tasdemir *et al* (2005).

The primary anthropogenic source of nitrogen oxides are the motor vehicles. The high value obtained at the commercial site which is also a busy traffic intersection vindicates it. The other sources are industries and diesel generating sets.

Temporal SOx concentration levels showed a similar trend. The concentration during the winter season was high as compared to the summer season. The minimum value in Kantatoli was recorded in August (14.34µg/m^3) while the maximum in December (30.69 µg/m^3). In Ashoknagar, the minimum concentration was seen in June (7.27 µg/m^3), while the maximum was recorded in May (25.26 µg/m^3). In Tupudana Industrial Area, the minimum values were obtained in August (12.97 µg/m^3) and the maximum in December (33.87 µg/m^3) .Table 1 shows the monthly variation in SOx values at three sampling sites. The high value of SOx at Ashoknagar during summer months can be attributed to some aberrant conditions that

might have prevailed there on the date of sampling. The primary anthropogenic source for SOx is the burning of fossil fuels.

Table 1: Monthly concentration of SOx and NOx in 2010-11 at select sampling sites

Month	Air Temp ^{0}C	Kantatoli (Commercial)		Ashoknagar (Residential)		Tupudana (Industrial)	
		SOx	NOx	SOx	NOx	SOx	NOx
December	16.6	30.69	64.8	15.16	21.02	33.87	59.39
January	17.4	27.57	51.03	11.87	26.93	29.07	42.69
February	18.3	27.78	55.05	14.36	37.16	21.03	34.15
March	21.5	25.24	43.78	20.18	23.02	17.14	47.12
April	30.6	26.76	46.81	16.09	25.71	25.67	34.34
May	31.3	25.67	43.91	25.26	34.08	28.22	37.16
June	30.6	28.25	43.78	7.27	22.68	28.25	29.32
July	26.1	21.48	22.85	12.75	24.52	16.84	39.13
August	24.3	14.34	36.13	10.39	16.4	12.97	33.82
September	28.6	16.87	38.2	21.2	26.48	25.04	43.63
October	21.8	28.04	55.58	12.83	21.07	27.57	41.03
November	20.9	27.84	55.87	14.36	22.68	29.9	46.81

CONCLUSION

The monthly sampling of SOx and NOx showed that there is a variation in the concentration of these two criteria air pollutants.The concentration of SOx and NOx is dependent on the season. During winter season the concentration increase while in summer the concentration decreases.

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